

5. (16%) Perturbations

How does SpaceX launch Starlink spacecraft into multiple orbital planes from a single launch without using fuel for plane changes?



DIFFERENTIAL $\dot{\Omega}_{J2}$

Five Starlink spacecraft from the latest launch will join the operational spacecraft in a plane that is currently 5° behind the launch plane. How long should these spacecraft wait to begin raising their orbit?

$$h_{park} = 440 \text{ km}$$

$$h_{msn} = 550 \text{ km}$$

$$i = 53^\circ$$

List your assumptions:

$$\dot{\Omega}_{J2} \gg \dot{\Omega}_\odot \text{ \& } \dot{\Omega}_\oplus, \text{ NO OTHER PERTURBATIONS AFFECT } \Omega$$

Solve:

$$\textcircled{1} \Delta\Omega = 5^\circ \times \frac{\pi}{180} = 0.08726 \text{ RAD}$$

$$\textcircled{2} \begin{cases} \dot{\Omega}_{J2}^{\text{PARK}} = -4.171 \times 10^{-7} \frac{\cos i}{a^{7/2}} = -4.171 \times 10^{-7} \frac{\cos 53^\circ}{6818^{7/2}} \\ \quad = -9.592 \times 10^{-7} \text{ RAD/S} \\ \dot{\Omega}_{J2}^{\text{MSN}} = -4.171 \times 10^{-7} \frac{\cos 53^\circ}{6928^{7/2}} \\ \quad = -9.069 \times 10^{-7} \text{ RAD/S} \end{cases}$$

$$a_{\text{PARK}} = 6378.137 + 440 \\ = 6818 \text{ km}$$

$$a_{\text{MSN}} = 6378.137 + 550 \\ = 6928 \text{ km}$$

$$\textcircled{3} \Delta\dot{\Omega}_{J2} = \dot{\Omega}_{J2}^{\text{MSN}} - \dot{\Omega}_{J2}^{\text{PARK}} = 5.225 \times 10^{-8} \text{ RAD/S}$$

$$\textcircled{4} \frac{5^\circ}{\Delta\dot{\Omega}_{J2}} = 1.670 \times 10^6 \text{ SEC} \\ = 19.33 \text{ DAYS}$$